Cheatsheet AZ-104 — Comandos: Azure CLI y PowerShell equivalentes, Cloud Shell y trucos de examen

1. Herramientas y sesión

Herramienta	Qué es	Detalles de examen
Azure portal	GUI; cada recurso tiene **Automation options** (exportar plantilla) y **Cloud Shell** integrado	Lo que pide "minimize administrative effort" para una sola acción
Azure CLI (`az`)	Multiplataforma (Windows, macOS, Linux, Docker); sintaxis `az <grupo> <subgrupo> <verbo> --opciones`; salida `-o json` (default), `-o table`, `-o tsv`; filtro **JMESPath** con `--query`	`az login` (`--use-device-code`, `--identity` en una VM) · `az account list -o table` · `az account set -s "Prod"` · `az account show` · `az configure --defaults group=RG1 location=westeurope` · `az upgrade` · `az find "vm create"` · `az interactive` · `az <cmd> --help`
Azure PowerShell (módulo **Az**)	Cmdlets `Verbo-AzSustantivo`; objetos por la tubería; PowerShell 7 en cualquier SO	`Install-Module Az -Scope CurrentUser` · `Update-Module Az` · `Connect-AzAccount` (`-UseDeviceAuthentication`, `-Identity`, `-Tenant`) · `Get-AzSubscription` · `Set-AzContext -Subscription "Prod"` · `Get-AzContext` · `Get-Command -Module Az.Compute -Verb Get` · `Get-Help New-AzVM -Examples`
Cloud Shell	Bash o PowerShell en el navegador (portal, `shell.azure.com`, app móvil, Windows Terminal, VS Code) con Az CLI, Az PowerShell, Bicep, Terraform, kubectl, git, editor (`code`); ya autenticado	Necesita una **storage account y un file share** (`clouddrive`, 5 GB) para persistir `$HOME` (opción **efímera** sin storage); se cierra tras **20 min** inactivo; 20 usuarios concurrentes por tenant; región del share fija
Módulos retirados	**AzureRM** (→ Az), **AzureAD** y **MSOnline** (→ **Microsoft Graph PowerShell**: `Connect-MgGraph`, `New-MgUser`, `Get-MgGroup`...)	En preguntas antiguas aparecen `New-AzureRmVM`, `New-AzureADUser`, `New-MsolUser`: la respuesta moderna es Az / Graph
Otros	`az rest` / `Invoke-AzRestMethod` (API REST sin SDK), SDKs, Azure Mobile app, Terraform, Bicep	"automate with code" → ARM / Bicep (cheatsheet 07)

2. Sintaxis que cae: JMESPath, pipeline y parámetros

```
az vm list -o table                                          # columnas básicas
az vm list -d --query "[?powerState=='VM running'].{Name:name, RG:resourceGroup}" -o table
az vm list --query "[?location=='westeurope'].name" -o tsv    # lista plana
az storage account list --query "[?sku.name=='Standard_LRS'] | length(@)"
az network nic show -g RG1 -n nic1 --query "ipConfigurations[0].privateIPAddress" -o tsv
az group list --query "[].{name:name, loc:location}" --output table
az vm create ... --no-wait                                   # no esperar; az vm wait --created
```

```
Get-AzVM -Status | Where-Object { $_.PowerState -eq "VM running" } | Select-Object Name, ResourceGroupName
Get-AzVM | Where-Object Location -eq "westeurope" |
    ForEach-Object { Stop-AzVM -Name $_.Name -ResourceGroupName $_.ResourceGroupName -Force -AsJob }
Get-AzResource -Tag @{Env="Prod"} | Select-Object Name, ResourceType
(Get-AzStorageAccount -ResourceGroupName RG1 -Name st1).PrimaryEndpoints.Blob
Get-AzRoleDefinition Contributor | ConvertTo-Json -Depth 5
Get-AzVM -ResourceGroupName RG1 -Name vm1 | Format-List *        # ver todas las propiedades
```

- Convenciones: CLI `-g` = `--resource-group`, `-n` = `--name`, `-l` = `--location`, `--ids` acepta IDs de recurso, `--yes` / `-y` sin confirmación, `--no-wait` asíncrono, JSON inline entre **comillas simples** en Bash (y escapado en PowerShell / cmd) o `@fichero.json` · PowerShell `-ResourceGroupName`, `-Name`, `-Location`, `-Force` sin confirmar, `-AsJob` en segundo plano (`Get-Job`, `Receive-Job`), `-WhatIf` simula, `-Confirm`, splatting `@params`.
- `New-` crea · `Set-` reemplaza / configura · `Update-` cambia propiedades concretas (`Update-AzVM`, `Update-AzTag`) · `Get-` lee · `Remove-` borra · `Add-` añade a una colección (`Add-AzVMDataDisk`) · `Start-` / `Stop-` / `Restart-`.

3. Identidad, RBAC y gobernanza

Tarea	Azure CLI	PowerShell
Usuarios y grupos de Entra	`az ad user create --display-name Ana --user-principal-name ana@contoso.com --password ...` · `az ad group create --display-name g1 --mail-nickname g1` · `az ad group member add --group g1 --member-id <id>` · `az ad user list --filter "startswith(displayName,'A')"`	`New-MgUser` · `New-MgGroup -SecurityEnabled -MailEnabled:$false -MailNickname g1 -DisplayName g1` · `New-MgGroupMember -GroupId <id> -DirectoryObjectId <id>` · `New-MgInvitation` · Az: `New-AzADUser`, `New-AzADGroup`, `Add-AzADGroupMember`
Licencias e identidades gestionadas	`az rest` (licencias) · `az identity create -g RG1 -n mi1` · `az vm identity assign -g RG1 -n vm1 [--identities mi1]`	`Set-MgUserLicense` · `New-AzUserAssignedIdentity` · `Update-AzVM -IdentityType SystemAssigned`
RBAC	`az role assignment create --assignee ana@contoso.com --role Contributor --resource-group RG1` · `az role assignment list --assignee ana@contoso.com --all` · `az role definition create --role-definition role.json`	`New-AzRoleAssignment -SignInName ana@contoso.com -RoleDefinitionName Contributor -ResourceGroupName RG1` · `Get-AzRoleAssignment -SignInName ... -ExpandPrincipalGroups` · `New-AzRoleDefinition -InputFile role.json`
Suscripciones y MGs	`az account management-group create -n mg1 --display-name "MG 1"` · `az account management-group subscription add -n mg1 --subscription <id>` · `az provider register --namespace Microsoft.Compute`	`New-AzManagementGroup -GroupName mg1` · `New-AzManagementGroupSubscription -GroupName mg1 -SubscriptionId <id>` · `Register-AzResourceProvider -ProviderNamespace Microsoft.Compute`

Tarea	Azure CLI	PowerShell
Resource groups	`az group create -n RG1 -l westeurope --tags Env=Prod` · `az group delete -n RG1 --yes --no-wait` · `az group list --tag Env=Prod` · `az group exists -n RG1`	`New-AzResourceGroup -Name RG1 -Location westeurope -Tag @{Env="Prod"}` · `Remove-AzResourceGroup -Name RG1 -Force -AsJob` · `Get-AzResourceGroup -Tag @{Env="Prod"}`
Tags	`az tag update --resource-id <id> --operation Merge --tags Owner=Ana` · `az resource tag --tags Env=Prod --ids <id>` · `az resource list --tag Env=Prod`	`Update-AzTag -ResourceId <id> -Tag @{Owner="Ana"} -Operation Merge` · `Set-AzResourceGroup -Name RG1 -Tag @{...}` · `Get-AzResource -Tag @{Env="Prod"}`
Locks	`az lock create -n L1 --lock-type CanNotDelete -g RG1 [--resource-name vm1 --resource-type Microsoft.Compute/virtualMachines]` · `az lock list` · `az lock delete`	`New-AzResourceLock -LockLevel CanNotDelete -LockName L1 -ResourceGroupName RG1 [-ResourceName vm1 -ResourceType ...]` · `Get-AzResourceLock` · `Remove-AzResourceLock -LockId <id>`
Policy	`az policy assignment create -n loc --policy <id o nombre> --scope <id> --params '{...}'` · `az policy state list --filter "complianceState eq 'NonCompliant'"` · `az policy remediation create`	`New-AzPolicyAssignment -Name loc -PolicyDefinition $def -Scope <id> -PolicyParameterObject @{...}` · `Get-AzPolicyState -Filter "ComplianceState eq 'NonCompliant'"` · `Start-AzPolicyRemediation`
Mover recursos	`az resource move --destination-group RG2 --ids <id1> <id2> [--destination-subscription-id <id>]`	`Move-AzResource -DestinationResourceGroupName RG2 -ResourceId <id1>,<id2>`
Presupuesto	`az consumption budget create --amount 500 --budget-name b1 --category cost --time-grain monthly --start-date ... --end-date ...`	`New-AzConsumptionBudget -Amount 500 -Name b1 -Category Cost -TimeGrain Monthly -StartDate ...`

4. Plantillas y despliegues

Tarea	Azure CLI	PowerShell
Desplegar en RG	`az deployment group create -g RG1 --template-file main.bicep --parameters main.bicepparam [--mode Complete] [--what-if]`	`New-AzResourceGroupDeployment -ResourceGroupName RG1 -TemplateFile main.bicep -TemplateParameterFile main.parameters.json [-Mode Complete] [-WhatIf]`
Suscripción / MG	`az deployment sub create -l westeurope --template-file main.bicep` · `az deployment mg create -m mg1 -l westeurope ...`	`New-AzSubscriptionDeployment -Location westeurope -TemplateFile main.bicep` · `New-AzManagementGroupDeployment`
Validar, historial, exportar	`az deployment group validate` · `az deployment group list -g RG1 -o table` · `az group export -n RG1` · `az deployment group export -g RG1 -n d1`	`Test-AzResourceGroupDeployment` · `Get-AzResourceGroupDeployment` · `Export-AzResourceGroup -ResourceGroupName RG1 -Path .` · `Save-AzResourceGroupDeploymentTemplate`
Bicep y template specs	`az bicep build / decompile / upgrade` · `az ts create -n ts1 --version 1.0 -g RG1 -l westeurope --template-file main.bicep`	`bicep build` · `New-AzTemplateSpec -Name ts1 -Version 1.0 -ResourceGroupName RG1 -Location westeurope -TemplateFile main.bicep`

5. Storage

Tarea	Azure CLI	PowerShell
Cuenta	`az storage account create -n st1 -g RG1 -l westeurope --sku Standard_GRS --kind StorageV2 --access-tier Hot --min-tls-version TLS1_2 --allow-blob-public-access false` · `az storage account update --sku Standard_ZRS` · `az storage account show -n st1 --query primaryEndpoints`	`New-AzStorageAccount -ResourceGroupName RG1 -Name st1 -Location westeurope -SkuName Standard_GRS -Kind StorageV2 -AccessTier Hot -MinimumTlsVersion TLS1_2` · `Set-AzStorageAccount ... -SkuName Standard_ZRS`
Claves y SAS	`az storage account keys list -n st1` · `az storage account keys renew --key primary` · `az storage account generate-sas --services b --resource-types sco --permissions rl --expiry ... --https-only` · `az storage container generate-sas -n c1 --permissions r --expiry ... --auth-mode login --as-user`	`Get-AzStorageAccountKey` · `New-AzStorageAccountKey -KeyName key1` · `New-AzStorageAccountSASToken -Service Blob -ResourceType Service,Container,Object -Permission rl -ExpiryTime ...` · `New-AzStorageContainerSASToken` · `New-AzStorageContainerStoredAccessPolicy`
Firewall	`az storage account update --default-action Deny --bypass AzureServices` · `az storage account network-rule add --vnet-name vnet1 --subnet sub1` · `--ip-address 203.0.113.0/24`	`Update-AzStorageAccountNetworkRuleSet -DefaultAction Deny -Bypass AzureServices` · `Add-AzStorageAccountNetworkRule -VirtualNetworkResourceId <id>` · `-IPAddressOrRange ...`
Blobs	`az storage container create -n c1 --account-name st1 --auth-mode login [--public-access blob]` · `az storage blob upload -f a.pdf -c c1 -n a.pdf --tier Cool` · `az storage blob set-tier --tier Archive` · `az storage blob undelete` · `az storage blob list` · `az storage blob download` · `az storage blob copy start`	`$ctx = New-AzStorageContext -StorageAccountName st1 -UseConnectedAccount` · `New-AzStorageContainer -Name c1 -Context $ctx -Permission Blob` · `Set-AzStorageBlobContent -File a.pdf -Container c1 -StandardBlobTier Cool -Context $ctx` · `Get-AzStorageBlob` · `Get-AzStorageBlobContent` · `Start-AzStorageBlobCopy`
Protección y lifecycle	`az storage account blob-service-properties update --enable-delete-retention true --delete-retention-days 14 --enable-versioning true --enable-container-delete-retention true` · `az storage account management-policy create --policy @policy.json`	`Enable-AzStorageBlobDeleteRetentionPolicy -RetentionDays 14` · `Enable-AzStorageContainerDeleteRetentionPolicy` · `Update-AzStorageBlobServiceProperty -IsVersioningEnabled $true` · `Set-AzStorageAccountManagementPolicy`
Files	`az storage share-rm create --storage-account st1 -n share1 --quota 1024 --enabled-protocols SMB` · `az storage file upload -s share1 --source f.txt` · `az storage share-rm snapshot` · `az storage account file-service-properties update --enable-delete-retention true --delete-retention-days 14`	`New-AzRmStorageShare -ResourceGroupName RG1 -StorageAccountName st1 -Name share1 -QuotaGiB 1024` · `Set-AzStorageFileContent` · `New-AzRmStorageShare -Snapshot` · `Update-AzStorageFileServiceProperty -EnableShareDeleteRetentionPolicy $true -ShareRetentionDays 14`
AzCopy	`azcopy login` · `azcopy copy "C:\d" "https://st1.blob.core.windows.net/c1?<SAS>" --recursive` · `azcopy sync ... --delete-destination=true` · `azcopy jobs resume <id>`	Igual (es una herramienta aparte)

6. Cómputo

Tarea	Azure CLI	PowerShell
VM	`az vm create -g RG1 -n vm1 --image Ubuntu2204 --size Standard_D2s_v5 --admin-username azureuser --generate-ssh-keys --zone 1 --vnet-name vnet1 --subnet sub1 --public-ip-address "" --nsg-rule NONE` · `az vm list -d -o table` · `az vm show -g RG1 -n vm1 -d --query publicIps`	`New-AzVM -ResourceGroupName RG1 -Name vm1 -Location westeurope -Image Win2022Datacenter -Size Standard_D2s_v5 -Credential (Get-Credential) -Zone 1 -VirtualNetworkName vnet1 -SubnetName sub1` · `Get-AzVM -Status` · `Get-AzPublicIpAddress`
Ciclo de vida	`az vm start / stop / deallocate / restart / redeploy / delete -g RG1 -n vm1` · `az vm resize --size Standard_D4s_v5` · `az vm list-vm-resize-options`	`Start-AzVM` · `Stop-AzVM -Force` (desasigna) · `Stop-AzVM -StayProvisioned` · `Restart-AzVM` · `Set-AzVM -Redeploy` · `$vm.HardwareProfile.VmSize = ...; Update-AzVM`
Discos	`az vm disk attach -g RG1 --vm-name vm1 --name d1 --new --size-gb 256 --sku Premium_LRS` · `az vm disk detach` · `az disk update -n d1 --size-gb 512` · `az snapshot create -g RG1 -n snap1 --source d1` · `az disk create --source snap1`	`Add-AzVMDataDisk -VM $vm -Name d1 -Lun 0 -CreateOption Empty -DiskSizeInGB 256 -StorageAccountType Premium_LRS; Update-AzVM` · `Remove-AzVMDataDisk` · `Update-AzDisk -ResourceGroupName RG1 -DiskName d1 -DiskUpdate (New-AzDiskUpdateConfig -DiskSizeGB 512)` · `New-AzSnapshot` · `New-AzDisk`
Extensiones y comandos	`az vm extension set -g RG1 --vm-name vm1 --name CustomScriptExtension --publisher Microsoft.Compute --settings '{...}'` · `az vm run-command invoke --command-id RunPowerShellScript --scripts "..."` · `az vm user update -u azureuser -p <pwd>` · `az vm boot-diagnostics enable`	`Set-AzVMCustomScriptExtension -FileUri ... -Run install.ps1 -Name cse ...` · `Invoke-AzVMRunCommand -CommandId RunPowerShellScript -ScriptPath fix.ps1` · `Set-AzVMAccessExtension` · `Set-AzVMBootDiagnostic -Enable`
Availability set, imágenes	`az vm availability-set create -n as1 -g RG1 --platform-fault-domain-count 2 --platform-update-domain-count 5` · `az vm generalize` · `az sig image-version create`	`New-AzAvailabilitySet -Sku Aligned -PlatformFaultDomainCount 2 -PlatformUpdateDomainCount 5` · `Set-AzVm -Generalized` · `New-AzGalleryImageVersion`
VMSS y autoscale	`az vmss create -g RG1 -n vmss1 --image Ubuntu2204 --orchestration-mode Flexible --instance-count 2 --zones 1 2 3 --lb lb1` · `az vmss scale --new-capacity 5` · `az vmss update-instances --instance-ids "*"` · `az monitor autoscale create --resource vmss1 --resource-type Microsoft.Compute/virtualMachineScaleSets --min-count 2 --max-count 10 --count 2` · `az monitor autoscale rule create --condition "Percentage CPU > 75 avg 10m" --scale out 2`	`New-AzVmss -OrchestrationMode Flexible -InstanceCount 2 -Zone 1,2,3 ...` · `Update-AzVmss -SkuCapacity 5` · `Update-AzVmssInstance -InstanceId "*"` · `New-AzAutoscaleScaleRuleObject` + `New-AzAutoscaleProfileObject` + `New-AzAutoscaleSetting`
App Service	`az appservice plan create -n plan1 -g RG1 --sku S1 --is-linux` · `az webapp create -n app1 -g RG1 --plan plan1 --runtime "DOTNETCORE:8.0"` · `az webapp deployment slot create --slot staging` · `az webapp deployment slot swap --slot staging --target-slot production` · `az webapp config appsettings set --settings K=v --slot-settings K` · `az webapp config hostname add --hostname www.contoso.com` · `az webapp config ssl bind --certificate-thumbprint <t> --ssl-type SNI` · `az webapp deploy --src-path app.zip` · `az webapp log tail`	`New-AzAppServicePlan -Tier Standard -NumberofWorkers 2 -WorkerSize Small ...` · `New-AzWebApp` · `New-AzWebAppSlot` · `Switch-AzWebAppSlot -SourceSlotName staging -DestinationSlotName production` · `Set-AzWebApp -AppSettings @{K="v"}` · `Set-AzWebAppSlotConfigName -AppSettingNames K` · `New-AzWebAppSSLBinding` · `Publish-AzWebApp -ArchivePath app.zip`
Contenedores	`az acr create -n acr1 -g RG1 --sku Premium` · `az acr build -r acr1 -t app:v1 .` · `az acr login -n acr1` · `az container create -g RG1 -n c1 --image acr1.azurecr.io/app:v1 --cpu 1 --memory 1.5 --dns-name-label app1 --ports 80 --restart-policy Never` · `az container logs` · `az containerapp env create` · `az containerapp create --min-replicas 0 --max-replicas 10 --ingress external --target-port 80`	`New-AzContainerRegistry -Sku Premium` · `Connect-AzContainerRegistry` · `New-AzContainerGroup -Container (New-AzContainerInstanceObject -Name c1 -Image ... -Port (New-AzContainerInstancePortObject -Port 80 -Protocol TCP)) -OsType Linux -IPAddressType Public -IPAddressDnsNameLabel app1 -Location westeurope` · `Get-AzContainerInstanceLog` · Container Apps: `New-AzContainerApp` (módulo Az.App)

7. Redes

Tarea	Azure CLI	PowerShell
VNet y subred	`az network vnet create -g RG1 -n vnet1 --address-prefixes 10.0.0.0/16 --subnet-name sub1 --subnet-prefixes 10.0.1.0/24` · `az network vnet subnet create / update` (`--network-security-group`, `--route-table`, `--service-endpoints`, `--delegations`, `--nat-gateway`, `--default-outbound false`) · `az network vnet update --dns-servers 10.0.0.4`	`New-AzVirtualNetwork -AddressPrefix 10.0.0.0/16 -Subnet (New-AzVirtualNetworkSubnetConfig -Name sub1 -AddressPrefix 10.0.1.0/24)` · `Add-AzVirtualNetworkSubnetConfig` / `Set-AzVirtualNetworkSubnetConfig` + `Set-AzVirtualNetwork` · `$vnet.DhcpOptions.DnsServers = @("10.0.0.4")`
IP, NIC	`az network public-ip create -n pip1 --sku Standard --allocation-method Static --zone 1 2 3 --dns-name app1` · `az network nic create --vnet-name vnet1 --subnet sub1 --public-ip-address pip1 --ip-forwarding true` · `az network nic ip-config update --private-ip-address 10.0.1.10`	`New-AzPublicIpAddress -Sku Standard -AllocationMethod Static -Zone 1,2,3` · `New-AzNetworkInterface -EnableIPForwarding` · `Set-AzNetworkInterfaceIpConfig -PrivateIpAddress 10.0.1.10; Set-AzNetworkInterface`
Peering	`az network vnet peering create -n hub-spoke --vnet-name hub --remote-vnet spoke --allow-vnet-access --allow-gateway-transit` (+ el inverso con `--use-remote-gateways --allow-forwarded-traffic`)	`Add-AzVirtualNetworkPeering -Name hub-spoke -VirtualNetwork $hub -RemoteVirtualNetworkId $spoke.Id -AllowGatewayTransit` (+ inverso `-UseRemoteGateways -AllowForwardedTraffic`)
NSG y ASG	`az network nsg create -n nsg1` · `az network nsg rule create --nsg-name nsg1 -n AllowRDP --priority 100 --direction Inbound --access Allow --protocol Tcp --source-address-prefixes 203.0.113.0/24 --destination-port-ranges 3389` · `az network nsg rule list --include-default` · `az network asg create` · `az network nic ip-config update --application-security-groups asg-web` · `az network nic list-effective-nsg`	`New-AzNetworkSecurityGroup -SecurityRules (New-AzNetworkSecurityRuleConfig -Name AllowRDP -Priority 100 -Direction Inbound -Access Allow -Protocol Tcp -SourceAddressPrefix 203.0.113.0/24 -SourcePortRange * -DestinationAddressPrefix * -DestinationPortRange 3389)` · `Add-AzNetworkSecurityRuleConfig; Set-AzNetworkSecurityGroup` · `New-AzApplicationSecurityGroup` · `Get-AzEffectiveNetworkSecurityGroup`
Rutas	`az network route-table create -n rt1 --disable-bgp-route-propagation true` · `az network route-table route create --route-table-name rt1 -n toNVA --address-prefix 0.0.0.0/0 --next-hop-type VirtualAppliance --next-hop-ip-address 10.0.100.4` · `az network nic show-effective-route-table`	`New-AzRouteTable -DisableBgpRoutePropagation` · `Add-AzRouteConfig -Name toNVA -AddressPrefix 0.0.0.0/0 -NextHopType VirtualAppliance -NextHopIpAddress 10.0.100.4; Set-AzRouteTable` · `Get-AzEffectiveRouteTable`

Tarea	Azure CLI	PowerShell
Bastion, NAT, endpoints	`az network bastion create -n bas1 --vnet-name vnet1 --public-ip-address pip-bas --sku Standard` · `az network nat gateway create -n nat1 --public-ip-addresses pip-nat` · `az network private-endpoint create --private-connection-resource-id <id> --group-id blob --connection-name c1 --vnet-name vnet1 --subnet sub1` · `az network private-dns zone create -n privatelink.blob.core.windows.net` · `az network private-dns link vnet create -z ... -n link1 -v vnet1 -e false`	`New-AzBastion -Sku Standard ...` · `New-AzNatGateway` · `New-AzPrivateEndpoint -PrivateLinkServiceConnection (New-AzPrivateLinkServiceConnection -PrivateLinkServiceId <id> -GroupId blob ...)` · `New-AzPrivateDnsZone` · `New-AzPrivateDnsVirtualNetworkLink`
DNS	`az network dns zone create -n contoso.com` · `az network dns record-set a add-record -z contoso.com -n www -a 203.0.113.10` · `az network dns record-set cname set-record -n app -c app1.azurewebsites.net` · `az network private-dns record-set a add-record`	`New-AzDnsZone` · `New-AzDnsRecordSet -Name www -RecordType A -Ttl 3600 -DnsRecords (New-AzDnsRecordConfig -IPv4Address 203.0.113.10)` · `New-AzPrivateDnsRecordSet`
Load Balancer	`az network lb create -n lb1 --sku Standard --public-ip-address pip-lb --frontend-ip-name fe --backend-pool-name be` · `az network lb probe create --protocol Http --port 80 --path /` · `az network lb rule create --protocol Tcp --frontend-port 80 --backend-port 80 --probe-name p80 --load-distribution SourceIP --floating-ip true` · `az network lb inbound-nat-rule create --frontend-port 50001 --backend-port 3389` · `az network nic ip-config address-pool add --lb-name lb1 --address-pool be`	`New-AzLoadBalancer -Sku Standard -FrontendIpConfiguration ... -BackendAddressPool ... -Probe (New-AzLoadBalancerProbeConfig -Protocol Http -Port 80 -RequestPath /) -LoadBalancingRule (New-AzLoadBalancerRuleConfig -Protocol Tcp -FrontendPort 80 -BackendPort 80 -LoadDistribution SourceIP -EnableFloatingIP)` · `Add-AzLoadBalancerInboundNatRuleConfig`
Diagnóstico	`az network watcher test-ip-flow --vm vm1 --direction Inbound --protocol TCP --local 10.0.1.4:3389 --remote 203.0.113.5:60000` · `az network watcher show-next-hop` · `az network watcher test-connectivity` · `az network watcher packet-capture create`	`Test-AzNetworkWatcherIPFlow` · `Get-AzNetworkWatcherNextHop` · `Test-AzNetworkWatcherConnectivity` · `New-AzNetworkWatcherPacketCapture`

8. Monitor y backup

Tarea	Azure CLI	PowerShell
Log Analytics	`az monitor log-analytics workspace create -n law1 -g RG1 --retention-time 90` · `az monitor log-analytics query -w <id> --analytics-query "..."` · `az monitor log-analytics workspace table update -n Event --retention-time 30 --total-retention-time 730`	`New-AzOperationalInsightsWorkspace -RetentionInDays 90` · `Invoke-AzOperationalInsightsQuery -WorkspaceId <id> -Query "..."` · `Update-AzOperationalInsightsTable`
Diagnostic settings y AMA	`az monitor diagnostic-settings create -n ds1 --resource <id> --workspace <id> --logs '[{"categoryGroup":"allLogs","enabled":true}]' --metrics '[{"category":"AllMetrics","enabled":true}]'` · `az vm extension set -n AzureMonitorWindowsAgent --publisher Microsoft.Azure.Monitor ...` · `az monitor data-collection rule association create`	`New-AzDiagnosticSetting -ResourceId <id> -WorkspaceId <id> -Log ... -Metric ...` · `Set-AzVMExtension -Publisher Microsoft.Azure.Monitor -ExtensionType AzureMonitorWindowsAgent ...` · `New-AzDataCollectionRuleAssociation`
Alertas	`az monitor action-group create -n ag1 --short-name ag1 --action email ops ops@contoso.com` · `az monitor metrics alert create -n cpu80 --scopes <id> --condition "avg Percentage CPU > 80" --window-size 5m --evaluation-frequency 1m --action <ag id>` · `az monitor scheduled-query create` · `az monitor activity-log alert create --condition category=Administrative and operationName=Microsoft.Compute/virtualMachines/delete`	`New-AzActionGroup -EmailReceiver (New-AzActionGroupEmailReceiverObject ...)` · `Add-AzMetricAlertRuleV2 -Condition (New-AzMetricAlertRuleV2Criteria ...)` · `New-AzScheduledQueryRule` · `New-AzActivityLogAlert`
Backup	`az backup vault create -n rsv1 -g RG1 -l westeurope` · `az backup vault backup-properties set --backup-storage-redundancy GeoRedundant --cross-region-restore-flag true` · `az backup protection enable-for-vm -v rsv1 -g RG1 --vm vm1 --policy-name DefaultPolicy` · `az backup protection backup-now` · `az backup recoverypoint list` · `az backup restore restore-disks --rp-name <rp> --storage-account st1 --target-resource-group RG2` · `az backup protection disable [--delete-backup-data true]` · `az backup job list`	`New-AzRecoveryServicesVault` · `Set-AzRecoveryServicesVaultContext` · `Set-AzRecoveryServicesBackupProperty -BackupStorageRedundancy GeoRedundant` · `Enable-AzRecoveryServicesBackupProtection -Policy $pol -Name vm1 -ResourceGroupName RG1` · `Backup-AzRecoveryServicesBackupItem` · `Get-AzRecoveryServicesBackupRecoveryPoint` · `Restore-AzRecoveryServicesBackupItem` · `Disable-AzRecoveryServicesBackupProtection [-RemoveRecoveryPoints]` · `Get-AzRecoveryServicesBackupJob`
Site Recovery	Sin comandos nativos (portal / PowerShell / REST)	`New-AzRecoveryServicesAsrPolicy -AzureToAzure -RecoveryPointRetentionInHours 24` · `New-AzRecoveryServicesAsrReplicationProtectedItem` · `Start-AzRecoveryServicesAsrTestFailoverJob` · `Start-AzRecoveryServicesAsrUnplannedFailoverJob` · `Start-AzRecoveryServicesAsrCommitFailoverJob`

9. Señales → respuesta

Enunciado	Respuesta
"run commands from any OS without installing anything"	Cloud Shell (necesita storage account + clouddrive, o modo efímero)
"Cloud Shell asks to create storage"	Primera vez: file share `clouddrive` para persistir `$HOME`
"Cloud Shell session ended"	Inactividad 20 min
"macOS admin must manage Azure with PowerShell"	PowerShell 7 + módulo Az (o Cloud Shell)
"New-AzureRmVM", "AzureRM"	Retirado: módulo Az (`New-AzVM`)
"New-AzureADUser", "New-MsolUser"	Retirados: Microsoft Graph PowerShell (`New-MgUser`)
"list VMs that are running"	`az vm list -d --query "[?powerState=='VM running']"` · `Get-AzVM -Status + Where-Object`

Enunciado	Respuesta
"only the private IP of the NIC"	`--query "...privateIPAddress" -o tsv` · `(Get-AzNetworkInterface ...).IpConfigurations[0].PrivateIpAddress`
"run the command without waiting"	`--no-wait` · `-AsJob`
"delete without confirmation"	`--yes` · `-Force`
"simulate before applying"	`--what-if` (deployments) · `-WhatIf`
"which command creates a role assignment"	`az role assignment create` · `New-AzRoleAssignment`
"deploy a Bicep file to a resource group"	`az deployment group create` · `New-AzResourceGroupDeployment`
"upload files to a container with identity"	`az storage blob upload --auth-mode login` · `Set-AzStorageBlobContent` con `New-AzStorageContext -UseConnectedAccount`
"stop the VM and stop billing"	`az vm deallocate` · `Stop-AzVM -Force`
"stop the VM but keep compute billing"	`az vm stop` · `Stop-AzVM -StayProvisioned`
"swap staging into production"	`az webapp deployment slot swap` · `Switch-AzWebAppSlot`
"enable backup for the VM"	`az backup protection enable-for-vm` · `Enable-AzRecoveryServicesBackupProtection`
"which NSG rule blocks the traffic"	`az network watcher test-ip-flow` · `Test-AzNetworkWatcherIPFlow`
"create the alert from the command line"	`az monitor metrics alert create` · `Add-AzMetricAlertRuleV2`

10. Errores a evitar

1. Confundir `az vm stop` (sigue facturando) con `az vm deallocate`, o `Stop-AzVM -StayProvisioned` con `Stop-AzVM -Force`.
2. Elegir cmdlets AzureRM, AzureAD o MSOnline en una pregunta nueva: Az y Microsoft Graph.
3. Olvidar `--auth-mode login` (o `-UseConnectedAccount`) y esperar que la identidad Entra funcione con los comandos de datos de storage.
4. Usar comillas dobles para JSON inline en Bash (`--settings '{"a":"b"}'` va con simples) o no escaparlas en PowerShell.
5. Creer que Cloud Shell guarda ficheros sin el file share `clouddrive` (modo efímero los pierde).
6. Pasar nombres donde el comando espera IDs (`--ids`, `-ResourceId`) o al revés.
7. Pensar que `az deployment group validate` o `Test-AzResourceGroupDeployment` comprueban cuotas y políticas.
8. Olvidar `-Force` / `--yes` en scripts desatendidos (se quedan esperando confirmación).